

Bill of Materials — Smartphone Battery finN-2e19

Document: VBF-T6R1F-BOM-01 | **Case:** t6_r1_flash | **Date:** 2026-08-25 | **Source:** cell/r5_finN_dfn_1c_calc.json (layer_kg_m2), LNMO.json

1. Layer Stack (single cell, area 0.1027 m²)

#	Layer	Material	Thickness (µm)	Porosity/loading	Density basis	kg/m²	Mass (g)
1	Positive current collector	Al foil	8	—	2700 kg/m³	0.02160	2.22
2	Positive electrode	LNMO spinel (LiNi $\blacksquare$. $\blacksquare$ Mn $\blacksquare$. $\blacksquare$ O $\blacksquare$)	74	ϵ_{act} 0.665	4120 kg/m³ (calc)	0.22121	22.72
3	Separator	PE/ceramic-coated	8	—	—	0.00168	0.17
4	Negative electrode	Graphite (dense-calendered)	120	ϵ_{act} 0.65	2260 kg/m³ (calc)	0.14913	15.32
5	Negative current collector	Cu foil	6	—	8960 kg/m³	0.05376	5.52
	Stack total		217.6			0.44739	45.95

Masses from calc-energy layer_kg_m2 × area. Electrolyte excluded from mass/volume (parameter set lacks density — contract caliber, `electrolyte\_included: false`).

2. Electrolyte (formulation estimate)

Component	Role	Parameter value
Base solvent	EC/EMC (3:7) + 1.2 M LiPF $\blacksquare$ (LNMO base)	—
High-transference additive	anion-immobilizing / single-ion-like additive	$t_{\blacksquare}$ = 0.9
Conductivity enhancer	fluorinated solvent + conductive additive	σ = 20 S/m
Diffusivity	Li $\blacksquare$ transport	D = 1e-9 m²/s
SEI modifier	LiF-rich SEI former (EC permeation block)	D_{ec} = 2e-19 m²/s

Transport/SEI values are design estimates (log `props\_source`: "estimate"); no electrolyte mass statement (parameter set lacks density).

3. Cell Totals (contract caliber, electrolyte/casing excluded)

Quantity	Value
Capacity (1C)	6.324 Ah
Energy	26.18 Wh
Mass	45.95 g

Quantity	Value
Volume (layer stack)	22.35 cm ³
ED volumetric	1171.7 Wh/L
ED gravimetric	569.9 Wh/kg
Thickness (stack)	217.6 μm

4. Thermal Management BOM (per cell)

Item	Specification	Effective h
Vapor chamber	0.4 mm Cu, in-plane K > 5000 W/mK	—
Graphite spreader	25 μm, in-plane K ~1500 W/mK	—
Aluminum frame contact	full-area, gap filler 1 W/mK	—
Total effective		300 W/m²K (h=250 verified minimum for M5)

5. Notes

- N/P ratio ≈ 1.15 (anode-to-cathode capacity incl. ϵ_{act} 0.65); chosen to keep the 4C charge real (~7 Ah) while minimizing the electrolyte-potential penalty that drives plating (see design_spec §4).
- All thickness/loading values are the `--params` file values of the final design (`\_params\_finN.json` + `EC diffusivity 2e-19`), simulation-verified.